

Day 9 – JavaScript: Variables, Data Types, and Operators

Objective:

Today, you'll begin with the basics of **JavaScript** — the programming language of the web. You'll understand how to **declare variables**, work with **data types**, and use **operators** to build logic.

1. What is JavaScript?

JavaScript is a **programming language** used to create interactive effects inside web browsers. It can **change content**, **validate forms**, **animate elements**, and more.

How to Link JavaScript to HTML

```
<script src="script.js"></script>
```

Or write directly inside:

```
<script>
  console.log("Hello from JS");
</script>
```

2. Variables in JavaScript

Variables store data values. You can declare variables using:

```
var name = "Dev";           // old way (avoid)
let age = 25;               // preferred
const pi = 3.14;           // constant value
```

3. Data Types

JavaScript supports different types of data:

Type	Example
String	"Hello"
Number	42, 3.14
Boolean	true, false
Array	[1, 2, 3]
Object	{name: "Dev"}
Undefined	let x;
Null	let y = null;

4. Operators

► Arithmetic Operators:

```
+ // Addition
- // Subtraction
* // Multiplication
/ // Division
% // Modulus (remainder)
```

► Assignment Operators:

```
= // assign
+= // x += 5 → x = x + 5
-= // x -= 2
```

► Comparison Operators:

```
== // equal to (value)
=== // equal to (value & type)
!= // not equal
> // greater than
< // less than
>= // greater than or equal to
```

► Logical Operators:

```
&& // AND
|| // OR
! // NOT
```

Example: Try in Console or Script

```
let name = "Dev";
let age = 22;
let isStudent = true;

console.log("Name:", name);
console.log("Age + 5 =", age + 5);
console.log("Is student?", isStudent);

let x = 10;
x += 5;
console.log("x =", x);

console.log(10 > 5); // true
```

Day 9 Task

Task Steps:

1. Create a file named `script.js`
 2. Declare variables for:
 - Your name, age, city
 - A boolean for student or not
 3. Perform arithmetic on numbers (add, subtract)
 4. Use comparison operators to check your age is greater than 18
 5. Log all values in the browser console
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★ Submission Instructions:

- ✉ Submit your file to: workdinarventure@gmail.com
- 📧 Subject: Day 9 - JavaScript Basics Task
- ! Don't just copy examples
- 🔍 Do small research and customize output
- ☐ Focus on writing clean and readable code